A Data-Driven Reinforcement Learning Enabled Battery Fast Charging Optimization Using Real-World Experimental Data

Jiarui He, Tianyi Yang, Ling Xie, Yikun Yang, Chunlin Chen, *Senior Member, IEEE*, and Jingwen Wei, *Member, IEEE*

***Abstract*—Fast and safe charging is of great significance for the application of lithium-ion batteries. As one of the most representative optimization decision-making algorithms, reinforcement learning is widely used in fast charging control. However, current research on fast charging based on reinforcement learning uses simulation data for optimization, which deviates from the actual battery. This paper proposes a fast charging strategy to optimize charging time and safety based on reinforcement learning. The agent is trained using the data from real experiment, which includes 600 charge-discharge cycles on each battery. To avoid getting trapped in locally optimal solution, we utilize the ϵ-greedy strategy and discuss the impact of different values of ϵ. Moreover, to avoid dangerous steps in real charging process, we explore the impact of simulating data to real data transfer on training results. We also conduct a comparison experiment to verify the superiority of our strategy. The results show that the strategy we proposed has advantages in terms of charging time, safety, and delaying battery aging. Besides, the transfer of simulation data to real data has been proven to be effective, which can avoid high-risk actions during the exploration process of real experiments and ensure the safety of the experiments.**

***Index Terms*—Electrochemical model (EM), fast charging, lithium-ion battery, parameter estimation, reinforcement learning, state traversal experiment.**

I. Introduction

THE increasing number of fuel-powered cars in the world has led to a shortage of fossil fuels and caused serious air pollution problems. Based on the advocacy of environment-friendly travel, the electrification of automobiles has become a new trend, and lithium-ion batteries are widely used in automotive power batteries due to their environmental friendliness and many other advantages [1]. As of now, compared to traditional gasoline-powered vehicles, there are two main issues with electric vehicles: the long time of charging and range anxiety. Therefore, optimizing the fast charging control of batteries is one of the key factors to promote the development of electric vehicles. During the battery charging process, simply increasing the charging current will increase the risk of thermal runaway and potentially accelerate battery aging [2], [3]. For the sake of safety and economic, it is crucial to propose a safe and efficient charging strategy.

To achieve the requirement of fast charging, a plethora of techniques applied to fast charging have been reported, which can be classified into the following four categories.

A. Model-Free

Model-free strategies employ predefined charging protocols that predetermine the values of charging current or voltage. The most common is the constant current–constant voltage (CCCV) charging strategy [4], which uses constant current charging in the early stages, and when the cutoff voltage is reached, it switches to constant voltage charging until the battery is fully charged. Another commonly used method is the multistage constant current (MCC) charging strategy [5], which sets gradually increasing threshold voltages for each stage and uses decreasing constant currents for charging. While the model-free strategies simplify the charging process, they ignore a series of internal reaction processes, which may cause serious safety issues during charging, accelerate the aging of batteries, and the potential of charging speed is not fully tapped.

B. Model Based

Model-based strategies build mathematical models that describe the dynamic behavior of batteries based on their behavioral characteristics. The model-based strategies can be divided into two categories: equivalent circuit models (ECMs) and electrochemical models (EMs). The ECM was originally proposed by Randles for rapid electrode reaction kinetics [6], and formed the foundation for modeling electrochemical energy storage systems. The ECM is a black box model, usually coupled with

Manuscript received 19 January 2024; revised 26 March 2024; accepted 22 April 2024. Date of publication 19 June 2024; date of current version 12 November 2024. This work was supported in part by the National Natural Science Fund of China under Grant 62203209. *(Corresponding author: Jingwen Wei.)*

The authors are with the School of Management and Engineering, Nanjing University, Nanjing 210093, China (e-mail: jrh@smail.nju.edu.cn; 522023150077@smail.nju.edu.cn; 522023150073@smail.nju.edu.cn; ykyang@smail.nju.edu.cn; clchen@nju.edu.cn; jwwei@nju.edu.cn).

Digital Object Identifier 10.1109/TIE.2024.3398687

0278-0046 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

the thermal model [7], that can roughly reflect the charging and discharging process of the battery and is widely used by researchers [8]. However, the ECM cannot fully describe a series of physical and chemical phenomena occurring inside lithium-ion batteries. During the complex charging process, the ECM may exhibit deviations in terms of battery voltage, temperature, and so on. Compared to ECM, EM is more like a white box model, which can truly reflect the internal physical and chemical reaction processes during the charging and discharging process of lithium-ion batteries, improving model accuracy and gradually gaining the favor of researchers [9]. The earliest EM is the Doyle–Fuller–Newman (DFN) model proposed by Newman and his collaborators [10], [11], [12]. Subsequently, simplified single-particle model (SPM) [13] and enhanced single-particle model (SPMe) [14] based on the DFN model are developed. Due to the high-precision advantage, many studies on lithium-ion batteries are conducted based on EM [15], [16], [17], [18], [19], [20]. Model-based algorithms have high-precision requirements for models, but higher precision leads to greater computational complexity.

C. Data-Driven

To overcome the challenge of model-based methods, some researchers have turned to data-driven methods. Data-driven methods leverage large amounts of historical data to learn the patterns in the charging process without requiring complex battery models. This approach has advantages in dealing with complex problems such as high dimensionality, nonlinearity, and time variability, and can effectively reduce computational complexity while improving the accuracy and robustness of charging strategies. Data-driven strategies bypass the battery modeling process and directly utilize existing massive data to train machine learning models that can predict battery life [21], based on this, a research team at Stanford University has proposed a closed-loop charging curve optimization method based on machine learning [22], which effectively extends the battery cycle life. Data-driven strategies are flexible and easier to implement, but require extensive time-consuming and energy-consuming testing experiments to generate training data, and the performance of the algorithm is highly dependent on the size and quantity of the data sets.

D. Learning Based

Learning-based methods usually utilize existing reinforcement learning methods in combination with lithium-ion battery simulation models to optimize the fast charging process [23], [24]. Compared to data-driven methods, the advantage of learning-based approaches is that the agent can achieve optimization goals through training and learning with relatively less data. The reinforcement learning algorithm learns by setting a reward mechanism and interacting with the environment. It can approach the completeness of knowledge even in the case of limited information and can effectively solve complex optimization problems. Since the training data come from a simulated environment, it inevitably leads to deviations from real batteries. If the training of agents is performed on real batteries, improper action selection may also lead to battery safety constraints being exceeded, resulting in unnecessary waste and increased risk during the training process.

For the sake of accuracy and feasibility, our research focuses on safe and fast charging based on reinforcement learning algorithms. The training data for the agent are derived from real battery experiments, and our optimization strategy is validated using EM. The main contributions are summarized as follows.

1) The DFN model is used as battery simulation model for the validation of optimization strategies. Before directly identifying the parameters of the DFN model, we reconstruct the electrodes of the battery by multiple consecutive fits to maximize the consistency between the model and the actual battery.
2) Unlike the aforementioned research work, the data for optimizing charging are based on the real experiment which traverses the states of the lithium-ion battery charging process to record the changes in battery voltage, temperature, and state of charge (SOC) when different currents are used in each state of the charging process. A unique reward mechanism is also used to avoid the voltage of the battery exceeding the threshold during the experiment process.
3) We use real battery data to train reinforcement learning agent, introduce the ϵ-greedy strategy, and combine the DFN model to investigate the impact of different values of ϵ on fast charging optimization results. Then, we explore the feasibility of the transfer process from simulation data to real data. The results show that compared to traditional charging strategy, the optimized charging strategy has advantages in charging time, charging safety, and delaying battery aging. In addition, the feasibility of data replacement has been verified, which can avoid some dangerous steps in the actual experimental process, improving the safety of the experiment.

II. Modeling and Algorithm

A. EM and Parameterization

Since the introduction of the DFN model, it has been widely used by researchers due to its high-precision modeling of lithium-ion batteries. Compared to other EMs, the DFN model is more comprehensive and can accurately describe the battery charging and discharging processes under complex conditions [25]. Therefore, we choose the higher precision DFN model as the simulation model.

In the DFN model, the battery is regarded as an assembly of multiple electrochemical reactors, each reactor consisting of multiple layers including electrolyte layer, electrode layer, and current collector layer. Each layer involves processes such as electrochemical reactions, ion diffusion, and charge transport. The electrode material of the DFN model consists of spherical particles, with a regular porous structure that avoids the complex structure and particle distribution of active materials in practice. Additionally, due to the high conductivity of the current collector, there is no significant change in the y direction and z direction, and only the electrochemical reaction kinetics

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

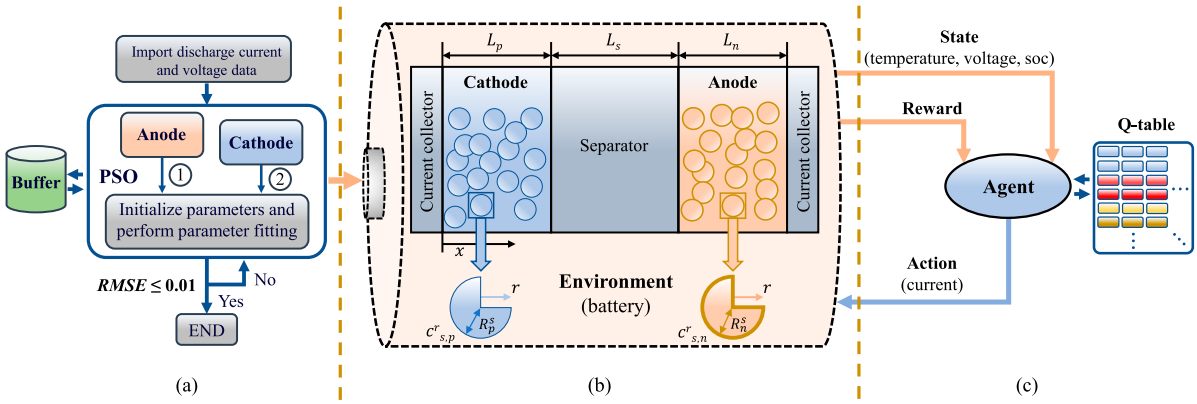


Fig. 1. (a) Flow chart for electrodes reconstruction. Use the PSO algorithm to fit the two electrode parameters multiple times in sequence, gradually reducing the RMSE. (b) Structure of the DFN model. L_p, L_s, and L_n represent the thicknesses of the cathode, separator, and anode, respectively. R_p^s, R_n^s, and c_s represent the particle radius of the cathode, particle radius of the anode, and lithium concentration in solid phase, respectively. (c) Coupling the reinforcement learning algorithm with the battery model.

in the x direction and radial direction r need to be considered. The structure of the DFN model is shown in Fig. 1(b). When the battery discharges, lithium in the anode particles becomes lithium ions, which move to the cathode through the separator and recombine in the cathode particles. The DFN model can be described by the following control equations [26], [27], [28].

Solid-phase diffusion equation

$$\frac{\partial c_s}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(D_s r^2 \frac{\partial c_s}{\partial r}\right). \tag{1}$$

Liquid-phase diffusion equation

$$\varepsilon_e \frac{\partial c_e}{\partial t} = \frac{\partial}{\partial x}\left(D_e^{\text{eff}} \frac{\partial c_e}{\partial x}\right) + a\left(1 - t_+^0\right) j. \tag{2}$$

Solid-phase charge conservation equation

$$\frac{\partial}{\partial x}\left(\sigma_s^{\text{eff}} \frac{\partial}{\partial x}\phi_s\right) = \text{Faj}. \tag{3}$$

Liquid-phase charge conservation equation

$$\frac{\partial}{\partial x}\left(\kappa_e^{\text{eff}} \frac{\partial \phi_e}{\partial x}\right) + \frac{\partial}{\partial x}\left(\kappa_D^{\text{eff}} \frac{\partial \ln c_e}{\partial x}\right) = -\text{Faj}. \tag{4}$$

The Butler–Volmer equation

$$\begin{cases} j = \frac{i_0}{F}\left[\exp\left(\frac{F}{2R_g T_{\text{cell}}}\eta_{ct}\right) - \exp\left(-\frac{F}{2R_g T_{\text{cell}}}\eta_{ct}\right)\right] \\ i_0 = k c_e^{0.5}\left(c_s^{\max} - c_s^{\text{surf}}\right)^{0.5}\left(c_s^{\text{surf}}\right)^{0.5}. \end{cases} \tag{5}$$

The variables in the equation are shown in Table I. To ensure the high fidelity of the DFN model, it is necessary to perform parameter identification on the model to make its charging and discharging characteristics consistent with the actual battery, thereby meeting the requirements of high-quality simulation. By using the charge and discharge data of real soft-pack batteries, we identify the parameters of the DFN model.

To ensure the accuracy of the subsequent fitting, the reconstruction of the battery electrodes is required before the parameters identification of the EM [29]. The open-circuit voltage (OCV) of the battery is determined by the positive and negative electrodes, as the final OCV is the potential difference between the two electrodes, as shown in

$$\text{OCV} = \text{OCV}_p - \text{OCV}_n. \tag{6}$$

Therefore, before fitting the OCV of the battery model, it is necessary to fit the OCV of both electrodes separately. The method we utilized is to fit the coefficients of the electrode by using the particle swarm optimization (PSO) algorithm [30], which has the advantages of fast convergence, few parameters,

TABLE I
NOMENCLATURE

Symbols	
a	Specific surface area of electrode (m^{-1})
c	Lithium-ion concentration (mol/m^3)
c_e	Electrolyte concentration (mol/m^3)
D	Diffusivity of electrolyte or solid phase (m^2/s)
D_s	Binary solid-phase diffusivity (m^2/s)
F	Faraday constant 96 485 C/mol
j	Lithium-ion pore wall flux [$\text{mol}/(\text{m}^2 \cdot \text{s})$]
R_g	Ideal gas constant 8.314 J/(kg · K)
t	Time (s)
t_0^+	Electrolyte transference number
T_{cell}	Battery temperature (K)
Greek Symbols	
ε	Volume fraction of electrolyte or solid phase
η_{ct}	Charge-transfer overpotential (V)
κ_D	Electrolyte diffusion conductivity (S/m)
κ_e	Electrolyte conductivity (S/m)
σ_s	Solid-phase conductivity (S/m)
ϕ	Potential of electrolyte or solid phase (V)
Subscripts & Superscripts	
e	Electrolyte phase
eff	Effective value
max	Maximum value
n	Anode
p	Cathode
s	Solid phase
surf	Surface value

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

and simple and easy implementation. If we only fit the two electrodes sequentially in a single iteration, it may result in an excessively large root mean square error (RMSE) that fails to meet the requirements. Therefore, we adopt a method of multiple consecutive fits as shown in Fig. 1(a). First, we fit the OCV of the anode, record the fitting coefficients in the buffer, which will be utilized to fit the OCV of the cathode, then fit the cathode. Repeat this process multiple times until the RMSE meets the requirement.

Subsequently, we utilize the control variable method to test multiple parameters of the DFN model and select the parameters with high sensitivity for identification based on the reconstructed electrodes. The training and validation conditions are dynamic stress test (DST) and urban dynamometer driving schedule (UDDS), respectively, and the PSO algorithm is also used for optimization. After iterative training, we update the identified parameters to the model to construct a high-fidelity electrochemical simulation model of lithium-ion batteries.

B. Problem Formulation and Q-Learning Algorithm

The charging process of lithium-ion batteries faces two problems: charging time and charging safety. The pursuit of charging speed will increase the risk of thermal runaway. Maintaining the battery temperature and voltage within a reasonable range is crucial for charging safety. The charging optimization problem makes a balance between charging time and charging safety, as showed

$$\min \quad J = J_{\text{time}} + J_{\text{safety}} \tag{7}$$

where J_{time} and J_{safety} denote charging time and battery safety loss, respectively. Our goal is to minimize loss. The constraints of the optimization problem are as follows:

$$\text{s.t.} \begin{cases} I_{\min} \le I(t) \le I_{\max} \\ V_{\min} \le V(t) \le V_{\max} \\ T_{\min} \le T(t) \le T_{\max} \\ 0 \le \text{SOC} \le \text{SOC}_{\text{goal}}. \end{cases} \tag{8}$$

The current $I(t)$ is the input variable. State variable $x(t) = [V(t), T(t), \text{SOC}]^\top$, include the voltage, temperature, and SOC at a specific time point. The constraints limit these variables within a certain range to ensure safety during the charging process. Under a certain battery state, selecting different charging currents will result in different changes in the state, which resemble the state-action-reward mechanism of reinforcement learning algorithms if a feedback is given. Therefore, by establishing an appropriate charging reward feedback mechanism, we can utilize reinforcement learning algorithms to solve the problem of fast charging optimization.

Q-learning algorithm is one of the most classic algorithms in reinforcement learning [31], which is well suited for the discretized state traversal experiments conducted here. Compared to other more complex reinforcement learning algorithms, Q-learning offers simplicity and ease of implementation. Q-learning algorithm is a value-based algorithm, which mainly includes five elements: agent, environment, state, action, and reward. Q represents a function $Q(s, a)$ that relates state s and action a and it is the expected reward that can be obtained by taking action a in a certain state s. The environment will provide a corresponding reward based on the agent's action feedback. The core idea of the algorithm is to construct a Q table to store the values of states and actions and then select the action that can obtain the maximum reward based on the Q value. As shown in Fig. 1(c), we couple the reinforcement learning with EM model. The environment is the charging and discharging process of the battery, the states are the temperature, SOC, and voltage of the battery, and the action is the charging current. The definition of reward mechanism is as follows:

$$\text{reward} = \begin{cases} -\lambda_1 \left(V(t) - V_{\max}\right), \text{ if } V(t) > V_{\max} \\ -\lambda_2 \left(T(t) - T_{\max}\right), \text{ if } T(t) > T_{\max} \\ 100, \text{ if } \text{SOC}(t-1) < \text{SOC}_{\text{goal}} \le \text{SOC}(t) \\ -2, \text{ other} \end{cases} \tag{9}$$

where λ_1 and λ_2 are reward coefficients. After charging for a certain period of time with a specific current in each state, we record the new state of the battery. If the voltage or temperature of battery exceeds the set threshold, a negative reward feedback will be given. If the SOC just meets the standard, a reward feedback of 100 will be given. In other cases, a time reward feedback of -2 will be given. Summarize the rewards corresponding to all states and actions into a Q table, and perform iterative training. Action selection based on the current state is actually a Markov process, and the optimal decision sequence for the Markov decision process can be solved through the Bellman equation

$$Q(s, a) \longleftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q\left(s', a'\right) - Q(s, a) \right] \tag{10}$$

where s' and a' denote the new state and next action, α and γ denote the learning rate and decay factor, and r denotes the reward. It is observed that the iterative formula in (10) always selects the action with the largest current reward in the selection of iterative actions, which may cause the results to fall into local optimal solution, as there may be other unexplored actions that have not been discovered due to their nonmaximum instantaneous reward values. Therefore, we introduce the ϵ-greedy strategy as shown in

$$\begin{cases} Q(s, a) \longleftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q\left(s', a'\right) - Q(s, a) \right] \\ \text{if } n \le \epsilon \\ Q(s, a) \longleftarrow Q(s, a) + \alpha \left[r + \gamma Q\left(s', a'\right) - Q(s, a) \right] \\ a' \text{ is random if } n > \epsilon. \end{cases} \tag{11}$$

The parameter n is a random number that follows a uniform distribution and ϵ is a preset constant, both of which are between 0 and 1. During the iteration process, there is a probability of $1 - \epsilon$ to choose a random action. The smaller the value of ϵ, the greater the probability of choosing a random action. The influence of different values of ϵ is discussed in Section IV.

The agent can make decisions based on the iterated Q table to find the optimal charging current sequence and achieve the goal of fast and safe charging. Therefore, the most important step in

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RESULT OF PARAMETER IDENTIFICATION

Parameters	value
Negative electrode diffusivity (m^2/s)	4.72119956e–14
Positive electrode diffusivity (m^2/s)	5.46172234e–15
Negative electrode active material volume fraction	7.61079183e–01
Positive electrode active material volume fraction	6.88452367e–01
Negative particle radius (m)	2.95514283e–06
Positive particle radius (m)	2.70029116e–06
Negative electrode porosity	3.11522067e–01
Positive electrode porosity	2.65178279e–01
Negative electrode electrons in reaction	1.49034782e + 00
Positive electrode electrons in reaction	1.61373852e + 00

implementing the Q-learning algorithm is the creation of the Q table. For this article, the Q table records the new state and reward feedback of the battery after selecting different currents for charging under different battery states. For the sake of accuracy, we conduct actual battery state traversal experiments to achieve this step.

III. EXPERIMENTS

The key to combining Q-learning algorithm with fast battery charging is to traverse the various states of the battery charging process. In this section, a real experiment on the traversal process will be performed instead of the simulation. The real data will be used to train the Q table to obtain more realistic optimization results.

The battery we used is 92103148-3.7-V ternary soft-pack lithium-ion battery, which has good charging and discharging characteristics and can withstand high currents. The experimental equipment is shown in Fig. 3(a). Upper computer controls the battery tester (NEWARE BTS4000) to conduct battery experiments and collects and records the battery data feedback from the tester for subsequent research. The thermal chamber (NEWARE MGDW-225-40) ensures that the battery is tested in a stable temperature environment, which guarantees the safety of the experiment and the accuracy of the data.

We reconstruct the electrodes of the EM to fit the OCV of the real battery. The fitted OCV is shown in Fig. 3(b), with a RMSE of 0.0093 V compared to the actual OCV. Subsequently, we use the DST step to identify the highly sensitive parameters of the model. After 30 iterations of the PSO algorithm, the training result of parameters identification is shown in Fig. 4(a), and the validation result of the UDDS step (the first 80%) is shown in Fig. 4(b). The RMSE of the training set and the validation set is 0.014 and 0.023 V, respectively. The parameter identification results are shown in Table II.

The design of the battery state traversal experiment is shown in Fig. 2: the state of the battery during charging is divided according to the SOC and temperature, as the SOC ranges from 0% to 95%, with each interval of 5% divided into a range, and the temperature ranges from 25 to 45 °C, with each interval of 5 °C divided into a range. So the battery charging process is divided into 100 states. The charging current is selected from 0.5 to 3 C at intervals of 0.5 C, with a total of 6 options. To ensure the matching between the state space and the action space, which means that after the agent performs an action, the new state it reaches should be within the defined set of state spaces, we charge the battery with certain current for 6 min at each state. Therefore, each battery in the experimental process needs to undergo 600 cycles. Considering that if charging the battery at a high SOC, the voltage may exceed the safety threshold, which will lead to dangerous accidents, so that the cutoff voltage set during the constant current charging process in the experiment is 4.2 V to ensure the safety of the experiment. In each cycle, we charge the battery to a certain SOC and set the temperature of the thermal chamber to a battery state that we will traverse, then select one of the six currents for constant current charging until it reaches 6 min or the battery reaches the cutoff voltage of 4.2 V. Record the charging time and the temperature of the battery, then discharge the battery and leave it to rest for 1 h. To shorten the time of the experiment, at certain moments, such as when the battery has undergone a 6-min constant current charge, the battery's SOC is exactly the state we will traverse next, allowing us to directly rest it for 1 h and perform the next cycle of experiments.

Due to the cutoff voltage setting, to achieve the same reward effect as before, we set the reward based on the charging time with constant current. The shorter the charging time, the greater the negative reward given. Therefore, the new reward mechanism is set as follows:

$$\text{reward} = \begin{cases} -\lambda_1 (6 - t_i), \text{ if } t_i < 6 \\ -\lambda_2 (T_i - T_{\max}), \text{ if } T_i > T_{\max} \\ 100, \text{ if } \text{SOC}_{i-1} < \text{SOC}_{\text{goal}} \leq \text{SOC}_i \\ -2, \text{ other} \end{cases} \tag{12}$$

where t denotes the charging time and i denotes the ith state. To facilitate the setting of reward mechanism, we set SOC_{goal} to 90%. When the charging capacity reaches SOC_{goal}, the optimization of the charging process is complete. The result visualization is shown in Fig. 2, where each square represents a state-action pair. Red indicates that the reward value is negative. Gold indicates that the reward value is positive. Blue indicates other situations in (12). The darker the color, the greater the absolute value of the reward.

As can be seen intuitively from results in Fig. 2, when the battery is in the same state (same SOC and temperature), the larger the current selected, the significantly increased negative reward value given (new SOC has not reached the target SOC). Because charging with a larger current will lead to a rapid rise in voltage until the cutoff voltage, shortening the charging time. According to (12), the negative reward value given at this case becomes larger. When charging with the same current at the same SOC, the increase in temperature will actually slightly weaken the negative reward, because the increase in temperature reduces the internal resistance of the battery, causing the terminal voltage of the battery to decrease accordingly which extends the time to reach the cutoff voltage. At the same time, it will also reduce the heat generated by the battery. According to (12), the negative reward value will be reduced.

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

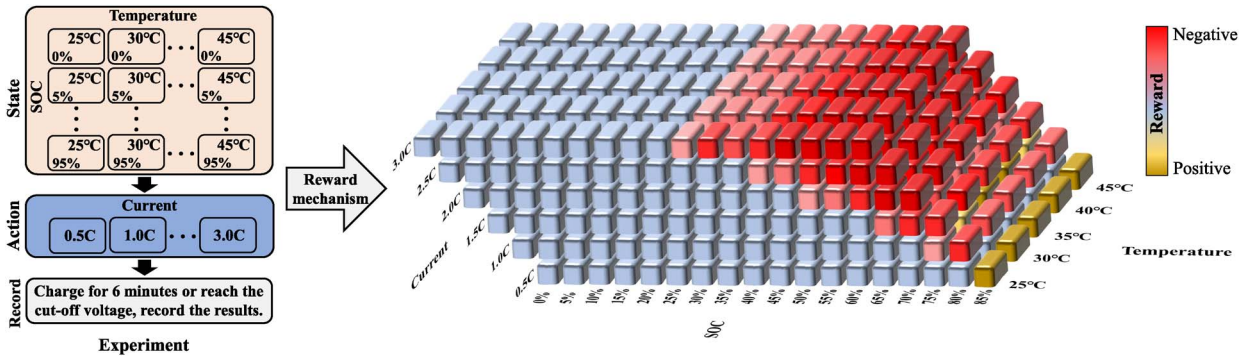
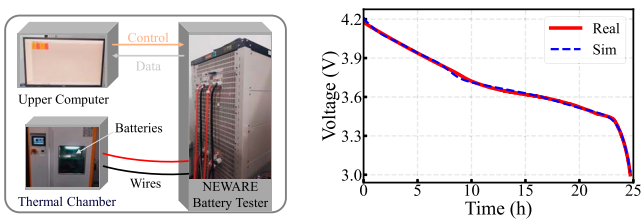
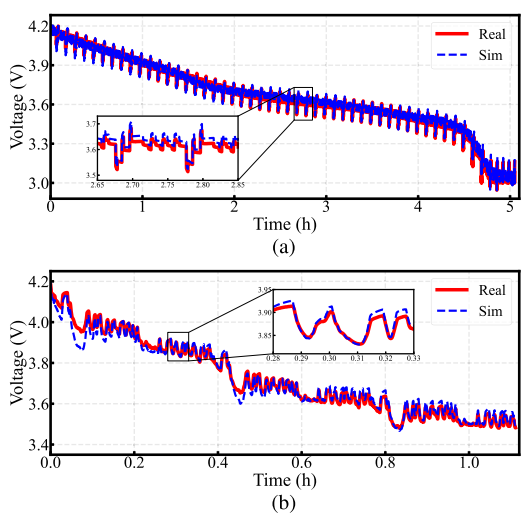


Fig. 2. Schematic diagram of states traversal experiment and results. Each square represents a state-action pair. Red indicates that the reward value is negative. Gold indicates that the reward value is positive. Blue indicates other situations. The darker the color, the greater the absolute value of the reward.



(a) (b)

Fig. 3. (a) Experiment equipment includes a computer control center, a thermostat, and a NEWARE battery tester. (b) Fitting result of OCV, RMSE = 0.0093 V.



(a) (b)

Fig. 4. (a) DST is used for the training set, and the RMSE is 0.014 V. (b) UDDS is used for the validation set, and the RMSE (the first 80%) is 0.023 V.

IV. Results and Discussion

The results of the actual state traversal experiment, combined with the reward feedback mechanism in (12), have completed the generation of the Q table. Then, we train the Q table with the iterative formula in (11). The values of ϵ we set are [1.0, 0.9, 0.8, 0.7], and other variables remain the same. The values of the parameters in (11) are shown in Table III.

TABLE III
Values of the Parameters

Parameters	Value	Parameters	Value
α	0.8	γ	0.8
λ_1	2	λ_2	5
$T_{\max}$	50 °C	$\mathrm{SOC}_{\mathrm{goal}}$	90%

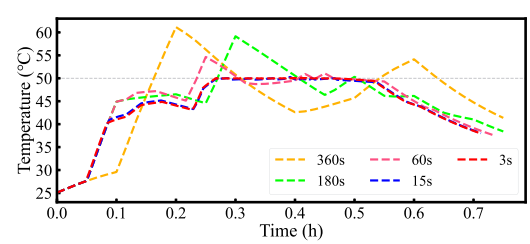


Fig. 5. Influence of different decision-making times of the agent on the battery temperature during the optimized charging process.

First, we investigate the decision-making time of the agent when optimizing fast charging. As shown in Fig. 5, under the condition of ϵ equaling 1.0, the set decision-making times are [360, 180, 60, 15, 3 s]. Observing the changes in battery temperature during the optimized charging process, it can be seen that as the decision-making time decreases, the precision of the charging process also becomes finer. The agent can judge the current battery state and select the corresponding charging current at a higher frequency, thus ensuring controllable battery conditions.

Fig. 6 demonstrates the impact of different values of ϵ on the optimized charging process. The interval between each decision-making of the agent is set to 3 s. As ϵ decreases from 1.0 to 0.7, the charging time are [2565, 2478, 2517, 2706 s]. When ϵ equals to 1.0 and 0.7, the charging time is longer than when it is equal to 0.9 and 0.8, with 1.0 representing the most

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

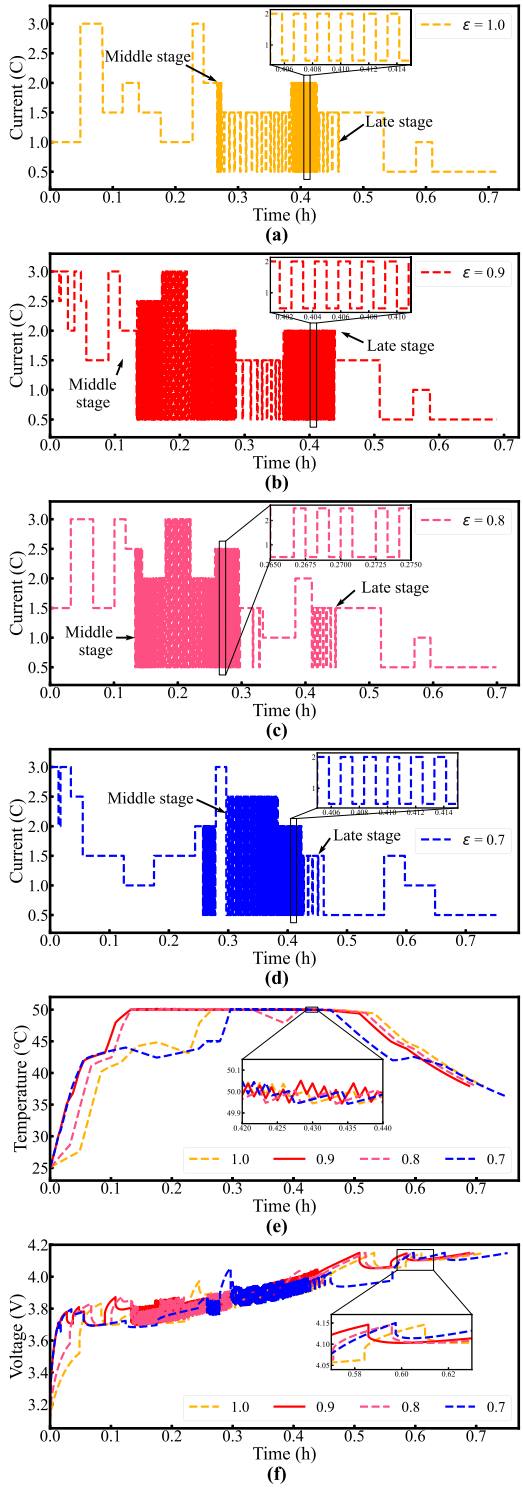


Fig. 6. (a)–(d) Optimized charging current corresponding to different ϵ. (e) Temperature change during charging process. (f) Voltage change during charging process.

conservative charging strategy and 0.7 representing the most radical charging strategy.

For the convenience of analysis, we divide the charging process into three stages based on the changes of temperature in Fig. 6(e).

1) Early stage: The battery temperature gradually changes to the threshold temperature.
2) Middle stage: The battery temperature remains near the threshold temperature.
3) Late stage: The battery temperature gradually drops until the end of charging.

From the charging current shown in Fig. 6, it can be seen that during the early stage, the charging processes are based on several constant current charging. When ϵ equals to 1.0, the temperature rise fluctuates, because the agent is in a conservative strategy. As when ϵ equals to 0.9 and 0.8, the temperature continues to rise to the threshold temperature, which represents the radical strategy. However, when ϵ continues to decrease to 0.7, the early stage becomes longer, the charging behavior becomes unexplained, weakening the fast charging control ability of the agent. Therefore, the accuracy of the agent's selection of the optimal path is not sufficient. In the middle stage of charging, all four conditions exhibit a pulsed charging current. As can be seen from Fig. 6(e), temperature limitation plays a major limiting role during this stage. The pulsed current maintains the battery temperature near the threshold, maximizing the charging speed. In the late stage of charging, the overall charging currents of all conditions are in the form of a small constant current, which is similar to the MCC strategy. As can be seen from Fig. 6(f), whenever the battery voltage is about to reach the threshold voltage, the agent will reduce the charging current to ensure that the battery voltage remains within a safe range so that voltage limitation plays a major limiting role. Therefore, in terms of temperature control and voltage control, all four cases ensure the safety of the charging process, and when ϵ equals to 0.9, the charging time is the shortest.

In addition, we explore the impact of simulation data to real data transfer on training results. We adopt the optimization strategy when ϵ equals to 0.9. The transfer ratio ranges from 0.93% to 13.9%, with state-action pairs counts of 5, 15, 30, 50, and 75, respectively. For each group, we conduct ten random replacement tests, and the results are analyzed in Fig. 7(a). By scaling the reward mechanism, it can be seen that as the number of substitutions increases, the mean value of the reward decreases, and after exceeding 5.56%, the variance also becomes very large. The best performance is achieved when five pairs are replaced. Therefore, we replace five pairs with large negative reward values for verification. We conduct a contrast experiment to simulate the charging of a battery with CCCV strategy, set the current in the constant current stage to 2 C, the voltage in the constant voltage stage to 4.2 V, and the cutoff condition to a current less than 0.05 C. As shown in Fig. 7(b), the training results of replacing five pairs are also the same as previous optimization results in temperature control, and both are superior to the CCCV strategy in terms of temperature safety, as CCCV strategy has a maximum temperature of 60.21 °C during the charging process. Moreover, the charging time for replacing five pairs is also the shortest among the three, with 2400 s, followed by 2478 s when ϵ equals to 0.9, while the CCCV strategy takes 2782 s to charge the battery to 90%.

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

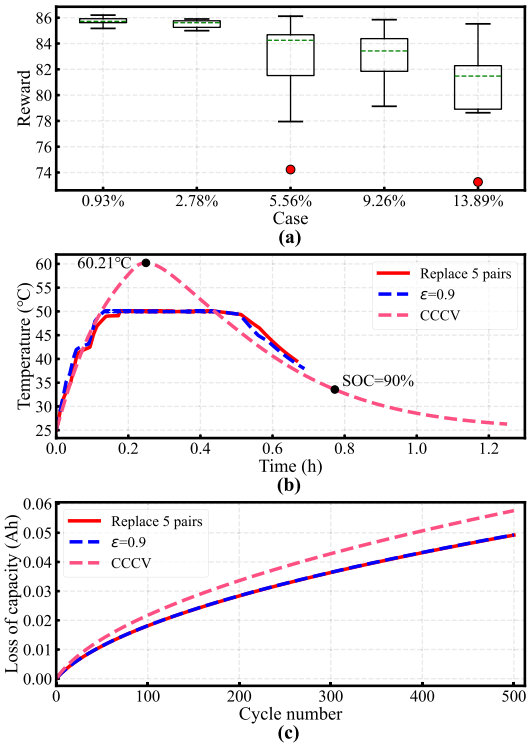


Fig. 7. (a) Results of replacement. (b) Temperature comparison results. (c) Aging experiment comparison results.

We also conduct aging analysis on the charging results, still using CCCV strategy as a comparison. In the simulation experiment, 500 cycles of charge and discharge are conducted, with the charging steps being the aforementioned CCCV strategy, the optimized strategy with ϵ equals to 0.9 and the replacing five pairs strategy, and the discharging process being performed with a constant current of 1 C. The results after 500 cycles are shown in Fig. 7(c). The capacity losses corresponding to the CCCV strategy, the strategy with ϵ equals to 0.9, and the strategy of replacing five pairs are 0.0576, 0.0493, and 0.0492 Ah, respectively, the replacement result once again shows the best performance. Therefore, the optimized strategy has advantages over the traditional CCCV strategy, whether in terms of charging time, temperature rise during charging, or battery aging.

V. Conclusion

Our article is based on reinforcement learning algorithm, utilizing battery state traversal experimental data to train the agent for solving the fast charging optimization problem and validating the optimized fast charging strategy with DFN model. First, we conducted parameter identification of the DFN model. After reconstructing the electrodes, we used the PSO algorithm combined with the DST step and UDDS step to train and validate parameter identification, obtaining a strictly simulated model that matches the real battery. Subsequently, experiments with real battery charging and discharging were conducted to traverse the states of the battery charging process, and based on the characteristics of the Q-learning algorithm, the Q table was trained to optimize charging time and safety. By ensuring that the temperature and voltage are within a safe range, fast charging of the battery can be achieved. To avoid getting trapped in local optima, we utilized the ϵ-greedy strategy and analyzed the impact of different values of ϵ on the optimization results. Then, we explored the transfer of simulation data to real data and compared it with the CCCV strategy and the original data training results. The results of transfer were superior to the CCCV strategy in terms of charging time, charging safety, and battery aging delay, and also slightly better than the original data training results. Therefore, the transfer of simulation data to real data were proven to be effective in optimizing the control of fast charging, which also avoids high-risk actions during the exploration process of real experiment and ensures the safety of the experiment.

References

[1] N. A. Chaturvedi, R. Klein, J. Christensen, J. Ahmed, and A. Kojic, "Algorithms for advanced battery-management systems," *IEEE Control Syst. Mag.*, vol. 30, no. 3, pp. 49–68, Jun. 2010.

[2] M. A. Hannan, M. H. Lipu, A. Hussain, and A. Mohamed, "A review of lithium-ion battery state of charge estimation and management system in electric vehicle applications: Challenges and recommendations," *Renewable Sustain. Energy Rev.*, vol. 78, pp. 834–854, Oct. 2017.

[3] P. Sun, R. Bisschop, H. Niu, and X. Huang, "A review of battery fires in electric vehicles," *Fire Technol.*, vol. 56, no. 4, pp. 1361–1410, Jan. 2020.

[4] S. S. Zhang, K. Xu, and T. Jow, "Study of the charging process of a LiCoO2-based li-ion battery," *J. Power Sour.*, vol. 160, no. 2, pp. 1349–1354, 2006.

[5] Y.-H. Liu and Y.-F. Luo, "Search for an optimal rapid-charging pattern for li-ion batteries using the Taguchi approach," *IEEE Trans. Ind. Electron.*, vol. 57, no. 12, pp. 3963–3971, Dec. 2010.

[6] J. E. B. Randles, "Kinetics of rapid electrode reactions," *Discuss. Faraday Soc.*, vol. 1, pp. 11–19, 1947, doi: 10.1039/DF9470100011.

[7] X. Hu, H. E. Perez, and S. J. Moura, "Battery charge control with an electro-thermal-aging coupling," in *Proc. Dynamic Syst. Control Conf.*, vol. 57243, New York, NY, USA: ASME, 2015, p. V001T13A002.

[8] J. Wei, G. Dong, and Z. Chen, "Remaining useful life prediction and state of health diagnosis for lithium-ion batteries using particle filter and support vector regression," *IEEE Trans. Ind. Electron.*, vol. 65, no. 7, pp. 5634–5643, Jul. 2018.

[9] S. K. Rahimian, S. Rayman, and R. E. White, "Comparison of single particle and equivalent circuit analog models for a lithium-ion cell," *J. Power Sour.*, vol. 196, no. 20, pp. 8450–8462, 2011.

[10] M. Doyle, T. F. Fuller, and J. Newman, "Modeling of galvanostatic charge and discharge of the lithium/polymer/insertion cell," *J. Electrochem. Soc.*, vol. 140, no. 6, 1993, Art. no. 1526.

[11] T. F. Fuller, M. Doyle, and J. Newman, "Simulation and optimization of the dual lithium ion insertion cell," *J. Electrochem. Soc.*, vol. 141, no. 1, 1994, Art. no. 1.

[12] J. Newman and K. Thomas-Alyea, "8.3 models for electrode kinetics," in *Electrochemical System.*, Hoboken, NJ, USA: Wiley, 2012, pp. 207–225.

[13] S. Santhanagopalan, Q. Guo, P. Ramadass, and R. E. White, "Review of models for predicting the cycling performance of lithium ion batteries," *J. Power Sour.*, vol. 156, no. 2, pp. 620–628, 2006.

[14] S. J. Moura, F. B. Argomedo, R. Klein, A. Mirtabatabaei, and M. Krstic, "Battery state estimation for a single particle model with electrolyte dynamics," *IEEE Trans. Control Syst. Technol.*, vol. 25, no. 2, pp. 453–468, Mar. 2017.

[15] L. D. Couto et al., "Faster and healthier charging of lithium-ion batteries via constrained feedback control," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 5, pp. 1990–2001, Sep. 2022.

[16] Y. Gao et al., "Health-aware multiobjective optimal charging strategy with coupled electrochemical-thermal-aging model for lithium-ion battery," *IEEE Trans. Ind. Inform.*, vol. 16, no. 5, pp. 3417–3429, May 2020.

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.

[17] C. Zou, X. Hu, Z. Wei, T. Wik, and B. Egardt, "Electrochemical estimation and control for lithium-ion battery health-aware fast charging," *IEEE Trans. Ind. Electron.*, vol. 65, no. 8, pp. 6635–6645, Aug. 2018.
[18] H. Ruan, B. Sun, W. Zhang, X. Su, and X. He, "Quantitative analysis of performance decrease and fast-charging limitation for lithium-ion batteries at low temperature based on the electrochemical model," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 1, pp. 640–650, Jan. 2021.
[19] Y. Li, T. Wik, Y. Huang, and C. Zou, "Nonlinear model inversion-based output tracking control for battery fast charging," *IEEE Trans. Control Syst. Technol.*, vol. 32, no. 1, pp. 225–240, Jan. 2024.
[20] Y. Li, Z. Wei, C. Xie, and D. M. Vilathgamuwa, "Physics-based model predictive control for power capability estimation of lithium-ion batteries," *IEEE Trans. Ind. Inform.*, vol. 19, no. 11, pp. 10763–10774, Nov. 2023.
[21] K. A. Severson et al., "Data-driven prediction of battery cycle life before capacity degradation," *Nature Energy*, vol. 4, no. 5, pp. 383–391, 2019.
[22] P. M. Attia et al., "Closed-loop optimization of fast-charging protocols for batteries with machine learning," *Nature*, vol. 578, no. 7795, pp. 397–402, 2020.
[23] Z. Wei, Z. Quan, J. Wu, Y. Li, J. Pou, and H. Zhong, "Deep deterministic policy gradient-DRL enabled multiphysics-constrained fast charging of lithium-ion battery," *IEEE Trans. Ind. Electron.*, vol. 69, no. 3, pp. 2588–2598, Mar. 2022.
[24] S. Park et al., "A deep reinforcement learning framework for fast charging of li-ion batteries," *IEEE Trans. Transp. Electrific.*, vol. 8, no. 2, pp. 2770–2784, Jun. 2022.
[25] S. Santhanagopalan, Q. Guo, P. Ramadass, and R. E. White, "Review of models for predicting the cycling performance of lithium ion batteries," *J. Power Sour.*, vol. 156, no. 2, pp. 620–628, 2006.
[26] M. Guo, G.-H. Kim, and R. E. White, "A three-dimensional multi-physics model for a Li-ion battery," *J. Power Sour.*, vol. 240, pp. 80–94, Oct. 2013.
[27] C. Li, N. Cui, C. Wang, and C. Zhang, "Simplified electrochemical lithium-ion battery model with variable solid-phase diffusion and parameter identification over wide temperature range," *J. Power Sour.*, vol. 497, 2021, Art. no. 229900.
[28] J. Kim, H. Chun, J. Baek, and S. Han, "Parameter identification of lithium-ion battery pseudo-2-dimensional models using genetic algorithm and neural network cooperative optimization," *J. Energy Storage*, vol. 45, 2022, Art. no. 103571.
[29] Z. Cui, N. Cui, C. Li, J. Lu, and C. Zhang, "Online identification and reconstruction of open-circuit voltage for capacity and electrode aging estimation of lithium-ion batteries," *IEEE Trans. Ind. Electron.*, vol. 70, no. 5, pp. 4716–4726, May 2023.
[30] J. Kennedy, "Particle swarm optimization," in *Proc. IEEE Int. Conf. Neural Netw.*, Perth, Australia, vol. 4, no. 8, 2011, pp. 1942–1948.
[31] C. J. Watkins and P. Dayan, "Q-learning," *Mach. Learn.*, vol. 8, pp. 279–292, May 1992.

Jiarui He received the B.E. degree in automation, in 2022, from Nanjing University, Nanjing, China, where he is currently working toward the M.E. degree in electronic information with the Department of Control and System Engineering.

His research interests include the fast-charging control of lithium-ion battery and reinforcement learning.

Tianyi Yang received the B.E. degree in automation, in 2023, from Nanjing University, Nanjing, China, where he is currently working toward the M.E. degree in electronic information with the Department of Control and System Engineering.

His research interests include battery fast charging control and optimal scheduling.

Ling Xie received the B.E. degree in automation, in 2023, from Nanjing University, Nanjing, China, where he is currently working toward the M.E. degree in electronic information with the Department of Control and System Engineering.

His research interest includes the fault diagnosis of lithium-ion battery.

Yikun Yang received the B.E. degree in automation, in 2021, from Nanjing University, Nanjing, China, where he is currently working toward the M.E. degree in electronic information with the Department of Control and System Engineering.

His research interests include battery management and reinforcement learning.

Chunlin Chen (Senior Member, IEEE) received the B.E. degree in automatic control and the Ph.D. degree in control science and engineering from the University of Science and Technology of China, Hefei, China, in 2001 and 2006, respectively.

He is currently a Professor and the Vice President of the School of Management and Engineering, Nanjing University, Nanjing, China. He was a Visiting Scholar with Princeton University, Princeton, NJ, USA, from 2012 to 2013. He had visiting positions at the University of New South Wales, Canberra, Australia, and the City University of Hong Kong, Hong Kong, China. His research interests include machine learning, pattern recognition, intelligent information processing, and quantum control.

Dr. Chen is the Chair of Technical Committee on Quantum Cybernetics, IEEE Systems, Man and Cybernetics Society. He is a Committee Member of JiangSu association of Artificial Intelligence (JSAI) Pattern Recognition Committee, and a Committee Member of China Association of Artificial Intelligence (CAAI) Machine Learning Committee.

Jingwen Wei (Member, IEEE) received the Ph.D. degree in control science and engineering from the University of Science and Technology of China, Hefei, China, in 2018.

She is currently an Assistant Professor with the Department of Control and Systems Engineering, Nanjing University, Nanjing, China. Her research interests include state estimation, balancing and health management, fault diagnosis, and fast charging control of lithium-ion batteries in electrical vehicles and energy storage.

Authorized licensed use limited to: Nanjing University. Downloaded on December 01,2025 at 12:36:47 UTC from IEEE Xplore. Restrictions apply.